

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12416935
Kind Code	B2
Date of Patent	September 16, 2025
Inventor(s)	Katsukura; Tomoya et al.

Control device and power conversion system

Abstract

A control device includes a modulation rate calculator configured to calculate a modulation rate based on a DC voltage on a DC side an, a limiter configured to update a DC voltage command lower bound limiter based on the lower bound limiter operation value; a maximum power point tracking controller configured to calculate a first DC voltage command value and determine a second DC voltage command value based on the calculated first DC voltage command value and the DC voltage command lower bound limiter set to the limiter, and an inverter controller configured to control an inverter of the power conversion system based on the second DC voltage command value.

Inventors:	Katsukura; Tomoya (Tokyo, JP), Tawada; Yoshihiro (Tokyo, JP), Fukasawa; Issei (Tokyo, JP), Kinoshita; Masahiro (Tokyo, JP)
Applicant:	TOSHIBA MITSUBISHI-ELECTRIC INDUSTRIAL SYSTEMS CORPORATION (Tokyo, JP)
Family ID:	1000008820997
Assignee:	TOSHIBA MITSUBISHI-ELECTRIC INDUSTRIAL SYSTEMS CORPORATION (Tokyo, JP)
Appl. No.:	18/700365
Filed (or PCT Filed):	October 05, 2022
PCT No.:	PCT/JP2022/037267
PCT Pub. No.:	WO2024/075212
PCT Pub. Date:	April 11, 2024

Prior Publication Data

Document Identifier	Publication Date
---------------------	------------------

Publication Classification

Int. Cl.: G05F1/67 (20060101); H02J3/38 (20060101); H02M7/5395 (20060101)

U.S. Cl.:

CPC G05F1/67 (20130101); H02J3/381 (20130101); H02M7/5395 (20130101); H02J2300/26 (20200101)

Field of Classification Search

CPC: G05F (1/67); H02J (3/38); H02J (3/381); H02J (2300/26); H02M (7/5395)

References Cited

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2018-195225	12/2017	JP	N/A

OTHER PUBLICATIONS

International Search Report & Written Opinion issued Dec. 20, 2022 in PCT/JP2022/037267, filed on Oct. 5, 2022, 10 pages (with English Translation). cited by applicant

Primary Examiner: DeBeradinis; Robert L

Attorney, Agent or Firm: Oblon, McClelland, Maier & Neustadt, L.L.P.

Background/Summary

FIELD

(1) The present invention relates to a control device and a power conversion system.

BACKGROUND

(2) Conventionally, a photovoltaics power conversion system configured to perform maximum power point tracking (MPPT) control has been known (refer to PTL 1, for example). Note that, in the present specification below, maximum power point tracking control is also referred to as “MPPT control” or simply as “MPPT”, and a maximum power point tracking controller configured to perform maximum power point tracking control (MPPT control) is also referred to as an “MPPT controller”.

CITATION LIST

Patent Literature

(3) [PTL 1] JP 2018-195225 A

SUMMARY

Technical Problem

(4) In such a power conversion system configured to perform MPPT control, MPPT control is constantly performed to track the maximum power point of power generated by a solar panel. In

MPPT control, information of DC power (DC current and DC voltage) is used to generate the next DC voltage command value, and thus information of AC voltage on the AC side (grid side) is not used.

(5) The power conversion system cannot stably operate at overmodulation, and thus the DC voltage command value needs to be provided with a lower bound limiter to avoid overmodulation. The lower bound limiter of DC voltage needs to be set with consideration on the operation range of grid voltage of the power conversion system, and the upper bound of the grid voltage needs to be substantially 110% of rated grid voltage as provided by grid interconnection regulations of each country.

(6) However, the grid voltage is near 100% for most of every day, and thus when the lower bound limiter of DC voltage is set to be higher to avoid overmodulation based on assumption of 110% of the grid voltage, the operation range on the DC side becomes narrow, which has been a problem.

(7) To solve this problem, it is conceivable to dynamically adjust the DC voltage command value by an MPPT controller in accordance with the grid voltage. However, as described above, MPPT control proceeds out of synchronization with state change of AC voltage. Accordingly, for example, when the DC voltage command value is updated based on a modulation rate during calculation of DC power by MPPT control, correct magnitude evaluation of DC power is potentially not performed by MPPT control. Thus, it has been conventionally impossible to generate the DC voltage command value with consideration on the modulation rate.

(8) In an energy-storage-system power conversion system, DC voltage is used to monitor the operation range of the power conversion system, and when the lower bound limiter of the DC voltage is variable in accordance with the modulation rate, a wider range can be set for the lower bound value of the voltage in accordance with the modulation rate and the operation range can be increased.

(9) Thus, the present disclosure is intended to provide a technology that can set a wider operation range on the DC side than in conventional cases by updating the lower bound limiter of DC voltage in accordance with the modulation rate and can stably operate a power conversion system without overmodulation.

Solution to Problem

(10) A control device according to an aspect is a control device in a photovoltaics power conversion system connected to a solar panel on a DC side and includes: a modulation rate calculator configured to calculate a modulation rate based on a DC voltage on a DC side of the power conversion system and a grid voltage root mean square value on an AC side of the power conversion system; a limiter adjuster configured to calculate a lower bound limiter operation value based on a predetermined lower bound limiter set value, the grid voltage root mean square value, and a predetermined rated value in accordance with the modulation rate calculated by the modulation rate calculator; a limiter configured to update a DC voltage command lower bound limiter based on the lower bound limiter operation value calculated by the limiter adjuster; a maximum power point tracking controller configured to calculate a first DC voltage command value by performing maximum power point tracking control based on DC power generated by the solar panel and determine a second DC voltage command value based on the calculated first DC voltage command value and the DC voltage command lower bound limiter set to the limiter; and an inverter controller configured to control an inverter of the power conversion system based on the second DC voltage command value determined by the maximum power point tracking controller.

(11) A photovoltaics power conversion system according to an aspect is connected to a solar panel at which power generation is performed by sunlight and includes an inverter configured to convert DC power generated by the solar panel into AC power, and the above-described control device.

(12) A control device according to another aspect is a control device in an energy-storage-system power conversion system connected to an energy storage system on a DC side and includes: a modulation rate calculator configured to calculate a modulation rate based on a DC voltage on a

DC side of the power conversion system and a grid voltage root mean square value on an AC side of the power conversion system; a limiter adjuster configured to calculate an operation range lower bound voltage operation value based on a predetermined operation range lower bound voltage set value, the grid voltage root mean square value, and a predetermined rated value in accordance with the modulation rate calculated by the modulation rate calculator; a limiter configured to update operation range lower bound voltage based on the operation range lower bound voltage operation value calculated by the limiter adjuster; a DC voltage operation range detector configured to compare the DC voltage on the DC side of the power conversion system and the operation range lower bound voltage set to the limiter and determine whether the DC voltage is smaller than the operation range lower bound voltage; and an operation controller configured to stop an inverter of the power conversion system when it is determined by the DC voltage operation range detector that the DC voltage is smaller than the operation range lower bound voltage.

(13) An energy-storage-system power conversion system according to another aspect is connected to an energy storage system capable of charging and discharging electric power and includes an inverter configured to convert DC power supplied from the energy storage system into AC power, and the above-described control device.

Advantageous Effects of Invention

(14) According to the present disclosure, it is possible to provide a technology that can set a wider operation range on the DC side than in conventional cases by updating the lower bound limiter of DC voltage in accordance with the modulation rate and can stably operate a power conversion system without overmodulation.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a diagram illustrating an example of the configuration of a control device and a power conversion system according to a first embodiment.

(2) FIG. 2 is a diagram illustrating an example of the configuration and processing of a modulation rate calculator and a limiter adjuster illustrated in FIG. 1.

(3) FIG. 3 is a flowchart illustrating an example of operation of an MPPT controller illustrated in FIG. 1.

(4) FIG. 4 is a diagram for description of an example of MPPT control based on a known hill climbing method by the MPPT controller illustrated in FIG. 1.

(5) FIG. 5 is a diagram for description of an example of specific processing in the operation of the MPPT controller described with reference to FIG. 3.

(6) FIG. 6 is a diagram illustrating an example of the configuration of a control device and a power conversion system according to a second embodiment.

(7) FIG. 7 is a diagram illustrating an example of the configuration and processing of a modulation rate calculator and a limiter adjuster illustrated in FIG. 6.

(8) FIG. 8 is a flowchart illustrating an example of operation of a DC voltage operation range detector and an operation controller illustrated in FIG. 6.

(9) FIG. 9 is a conceptual diagram illustrating an example of a hardware configuration of a processing circuit included in the control device in each embodiment illustrated in FIGS. 1 to 8.

DESCRIPTION OF EMBODIMENTS

(10) Embodiments of a control device and a power conversion system according to the present disclosure will be described below with reference to the accompanying drawings.

Configuration of First Embodiment

(11) FIG. 1 is a diagram illustrating an example of the configuration of a control device **30** and a power conversion system **1** according to a first embodiment. In FIG. 1, the power conversion

system **1** is connected to a solar panel **4** through a DC bus **2** on one end side (input side) on the left side in FIG. **1** and is connected to an AC power grid **5** through an AC circuit **3** on the other end side (output side) on the right side in FIG. **1**.

(12) For example, the power conversion system (PCS: Power Conditioning System) **1** converts DC power supplied from the solar panel **4** into AC power and outputs the converted AC power to the AC power grid **5** side (AC side). In other words, the power conversion system **1** in the present embodiment is a photovoltaics (PV) power conversion system (PV-PCS: Photovoltaics-Power Conditioning System). Hereinafter, in the present specification, the power conversion system **1** is also referred to as the “PV-PCS **1**” or simply as the “PCS **1**”.

(13) As illustrated in FIG. **1**, the power conversion system **1** includes a DC switch **11**, a DC capacitor **12**, an inverter **13**, an AC reactor **14**, an AC capacitor **15**, and an AC switch **16**. The power conversion system **1** also includes a first current sensor **21**, a first voltage sensor **22**, a second current sensor **23**, a second voltage sensor **24**, a third current sensor **25**, and the control device **30**. Note that the control device **30** is electrically connected to each elements of the power conversion system **1** although wires are omitted in the drawing.

(14) The DC bus **2** connects the solar panel **4** and a DC end (input side) of the inverter **13** of the power conversion system **1**. The DC bus **2** supplies DC power generated by the solar panel **4** to the inverter **13**. For example, the first current sensor **21**, the DC switch **11**, the first voltage sensor **22**, and the DC capacitor **12** are arranged on the DC bus **2** in order from the solar panel **4** toward the DC end of the inverter **13**.

(15) The AC circuit **3** connects an AC end (output side) of the inverter **13** of the power conversion system **1** and the AC power grid **5**. The AC circuit **3** is, for example, a three-phase three-wire three-phase AC circuit configured to supply, by using three electrical lines or cables, three-phase AC power as combination of three-grid single-phase AC currents having current or voltage phases shifted from one another. The AC circuit **3** supplies AC power converted by the inverter **13** to the AC power grid **5** side. For example, the second current sensor **23**, the AC reactor **14**, the AC capacitor **15**, the AC switch **16**, the second voltage sensor **24**, and the third current sensor **25** are arranged in the AC circuit **3** in order from the AC end of the inverter **13** toward the AC power grid **5**.

(16) The solar panel (solar battery panel) **4** is connected to the DC end (input end) of the inverter **13** through the DC bus **2**. The solar panel **4** is referred to as a solar module or a solar battery module or simply as a solar battery, a module, or the like and is, for example, one panel as combination of a plurality of solar battery cells. The solar panel **4** may be, for example, a solar battery string as combination of a plurality of panels in series or parallel to one another, or a solar battery array as combination of solar battery strings. The solar panel **4** generates power with sunlight and supplies generated DC power to the inverter **13** through the DC bus **2**. Note that, in the present specification below, the solar panel **4** is also referred to as the “solar battery panel **4**” or the “PV (photovoltaics) panel **4**”.

(17) The AC power grid (grid) **5** is connected to the AC end (output end) of the inverter **13** through the AC circuit **3**. The AC power grid **5** is an integrated system of power generation, power transformation, power transmission, and power distribution, which is connected to a non-illustrated transformer to supply AC power transformed by the non-illustrated transformer to a power receiving facility of a consumer, and is connected to, for example, an unspecified load. Note that, in the present specification below, the AC power grid **5** is also referred to as the “power grid **5**” or simply as the “grid **5**”.

(18) The DC switch (DC breaker) **11** is provided between the first current sensor **21** and the first voltage sensor **22** on the DC bus **2**. The DC switch **11** closes (connects) or opens (cuts) the DC bus **2** between the solar panel **4** and the inverter **13** in accordance with a close instruction or an open instruction from the control device **30**, a non-illustrated operator, or the like. When the DC switch **11** is opened, DC power supplied from the solar panel **4** to the inverter **13** is cut. Note that, in the

present specification below, the DC switch **11** is also referred to as the “DC switch **11**”.

(19) The DC capacitor **12** is provided between the first voltage sensor **22** and the DC end of the inverter **13** on the DC bus **2**. The DC capacitor **12** is a smoothing capacitor configured to smooth DC voltage output from the solar panel **4**. For example, the DC capacitor **12** is charged by DC power from the solar panel **4** and voltage thereof increases when the DC switch **11** is closed, and the DC capacitor **12** is discharged by a discharging circuit, a discharging resistance, or the like non-illustrated and voltage thereof decreases when the DC switch **11** is opened. Hereinafter, in the present specification, the DC capacitor **12** is also referred to as the “DC capacitor **12**”.

(20) In the inverter (inverter circuit) **13**, the one end side as the DC end is connected to the DC capacitor **12** and the DC switch **11** through the DC bus **2**, and the other end side as the AC end is connected to the AC reactor **14** through the AC circuit **3**. The inverter circuit **13** is constituted by a plurality of switching elements such as insulated gate bipolar transistors (IGBT). The inverter circuit **13** is controlled by, for example, a pulse width modulation (PWM) signal that is a gate drive signal (gate signal) generated for the switching elements by a pulse width modulation (PWM) controller **65** to be described later.

(21) The inverter **13** acquires, from the one end side, DC power supplied from the solar panel **4**, converts the acquired DC power into AC power in accordance with control by a pulse width modulation signal (gate signal), and supplies the AC power to the grid **5** side through outputting from the other end side as the output end. Note that, in the present specification below, the pulse width modulation signal is also referred to as the “PWM signal”.

(22) The AC reactor **14** is connected in series to the AC end of the inverter **13** in the AC circuit **3**. The AC reactor **14** is a smoothing element having, for example, a noise reducing effect and a surge voltage suppressing effect. For example, the AC reactor **14** constitutes, together with the AC capacitor **15** connected in an L shape through a bifurcation point **15a**, an LC filter circuit (filter circuit) configured to reduce ripples (vibration) that occur at switching of the non-illustrated switching elements of the inverter **13**. Note that, in the present specification below, the AC reactor **14** is also referred to as the “AC reactor **14**”.

(23) The AC capacitor **15** is an electronic component configured to charge or discharge electricity (electric charge) and is connected to the AC reactor **14** in an L shape through the bifurcation point **15a** in the AC circuit **3**. For example, the AC capacitor **15** constitutes, together with the AC reactor **14** connected in an L shape, the LC filter circuit (filter circuit) configured to reduce ripples (vibration) that occur at switching of the non-illustrated switching elements of the inverter **13**. The AC capacitor **15** constituting the filter circuit together with the AC reactor **14** reduces harmonic (harmonic current) outflow to a non-illustrated grid side. Hereinafter, in the present specification, the AC capacitor **15** is also referred to as the “AC capacitor **15**”.

(24) The AC switch (AC breaker) **16** is provided in series between the AC power grid **5** and each of the AC reactor and the AC capacitor in the AC circuit **3**. The AC switch **16** closes (connects) or opens (cuts off) the AC circuit **3** between the inverter **13** and the grid **5** in accordance with a close instruction or an open instruction from the control device **30**, the non-illustrated operator, or the like. AC power supplied from the inverter **13** to the grid **5** side is cut when the AC switch **16** is opened. Hereinafter, in the present specification, the AC switch **16** is also referred to as the “AC switch **16**”.

(25) The first current sensor **21** is, for example, a known DC ammeter or DC sensor, is arranged between the PV panel **4** and the DC switch **11** on the DC side, and detects the value of DC current $i_{\text{sub.DC}}$ flowing from the PV panel **4**. Note that the position at which the first current sensor **21** is arranged is not limited to the position illustrated in FIG. **1** but may be any position at which the first current sensor **21** can detect the value of the DC current $i_{\text{sub.DC}}$ flowing from the PV panel **4**. Hereinafter, in the present specification, the value of the DC current $i_{\text{sub.DC}}$ is also simply referred to as the “DC current $i_{\text{sub.DC}}$ ”. The DC current $i_{\text{sub.DC}}$ detected by the first current sensor **21** is acquired by the control device **30**.

(26) The first voltage sensor **22** is, for example, a known DC voltmeter or DC voltage sensor, is arranged between the DC switch **11** and the DC capacitor **12** on the DC side, and detects the value of DC voltage $v_{\text{sub.DC}}$ of the DC capacitor **12**. Note that the position at which the first voltage sensor **22** is arranged is not limited to the position illustrated in FIG. **1** but may be any position at which the first voltage sensor **22** can detect the value of the DC voltage $v_{\text{sub.DC}}$ of the DC capacitor **12**. Hereinafter, in the present specification, the value of the DC voltage $v_{\text{sub.DC}}$ is also simply referred to as the “DC voltage $v_{\text{sub.DC}}$ ”. The DC voltage $v_{\text{sub.DC}}$ detected by the first voltage sensor **22** is acquired by the control device **30**.

(27) The second current sensor **23** is, for example, a known AC ammeter or AC sensor, is arranged between the inverter **13** and the AC reactor **14** on the AC side, and detects the value of inverter output current $i_{\text{sub.AC}}$ as three-phase AC current output from the inverter **13**. Note that the position at which the second current sensor **23** is arranged is not limited to the position illustrated in FIG. **1** but may be any position at which the second current sensor **23** can detect the value of the inverter output current $i_{\text{sub.AC}}$ output from the inverter **13**. Hereinafter, in the present specification, the value of the inverter output current $i_{\text{sub.AC}}$ is also simply referred to as the “inverter output current $i_{\text{sub.AC}}$ ”. The inverter output current $i_{\text{sub.AC}}$ detected by the second current sensor **23** is acquired by the control device **30**.

(28) The second voltage sensor **24** is, for example, a known AC voltmeter or AC voltage sensor, is arranged between the AC switch **16** and the grid **5** on the AC side, and detects the value of grid voltage $v_{\text{sub.Grid}}$ as three-phase AC voltage in the grid **5**. Note that the position at which the second voltage sensor **24** is arranged is not limited to the position illustrated in FIG. **1** but may be any position at which the second voltage sensor **24** can detect the value of the grid voltage $v_{\text{sub.Grid}}$ in the grid **5**. Hereinafter, in the present specification, the value of the grid voltage $v_{\text{sub.Grid}}$ is also simply referred to as the “grid voltage $v_{\text{sub.Grid}}$ ”. The grid voltage $v_{\text{sub.Grid}}$ detected by the second voltage sensor **24** is acquired by the control device **30**.

(29) The third current sensor **25** is, for example, a known AC ammeter or AC sensor, is arranged between the AC switch **16** and the grid **5** on the AC side, and detects the value of AC current $i_{\text{sub.Grid}}$ as three-phase AC current flowing to the grid **5**. Note that the position at which the third current sensor **25** is arranged is not limited to the position illustrated in FIG. **1** but may be any position at which the third current sensor **25** can detect the value of the AC current $i_{\text{sub.Grid}}$ flowing to the grid **5**. Hereinafter, in the present specification, the value of the AC current $i_{\text{sub.Grid}}$ is also simply referred to as “grid current $i_{\text{sub.Grid}}$ ”. The grid current $i_{\text{sub.Grid}}$ detected by the third current sensor **25** is acquired by the control device **30**.

(30) For example, the control device **30** is provided inside or outside the power conversion system **1** and connected to each component of the power conversion system **1** such as the inverter **13** in a wired or wireless manner although wires and the like are omitted in the drawing. Note that the control device **30** may be implemented as a function of a non-illustrated inverter control circuit.

(31) The control device **30** includes a processor **91** (refer to FIG. **9**) to be described later, such as a central processing unit (CPU), a micro processing unit (MPU), or a graphics processing unit (GPU), the processor **91** being configured to operate by executing a computer program. The control device **30** also includes a storage **70** and a memory **92** (refer to FIG. **9**) to be described later and collectively controls operation of the power conversion system **1** by operating the processor **91** through, for example, execution of a predetermined computer program stored in the storage **70** or the memory **92**.

(32) Note that the control device **30** may operate in accordance with, for example, an instruction from a non-illustrated higher-level device or an instruction from the operator through a non-illustrated operation unit. Note that, for example, the non-illustrated higher-level device collectively monitors or controls a plurality of power conversion systems **1** and is connected to each power conversion system **1** in a wired or wireless manner.

(33) As illustrated in FIG. **1**, the control device **30** includes components or functions as a phase

locked loop (PLL) controller **41**, a first converter **42**, a second converter **43**, and a power controller **44**. The control device **30** also includes components or functions as a root mean square value (RMS) calculator **51**, a modulation rate calculator **52**, and a limiter adjuster **53**. The control device **30** also includes components or functions as a limiter **54**, a maximum power point tracking controller (MPPT controller) **55**, a first subtractor **56**, and a DC voltage controller **57**. Note that, in the present embodiment, the modulation rate calculator **52** includes a component or a function as the limiter adjuster **53**, and the MPPT controller **55** includes a component or a function as the limiter **54**. The control device **30** also includes components or functions as a first adder **61**, a third converter **62**, a second subtractor **63**, a current controller **64**, and a PWM controller **65**.

(34) Note that each above-described component or function may be implemented by a computer program executed by the processor **91** (refer to FIG. 9) to be described later in a processing circuit **90** (refer to FIG. 9) included in the control device **30** and to be described later or may be implemented by a hardware component **93** (refer to FIG. 9) to be described later. The control device **30** executes a predetermined computer program and performs processing below with each above-described component or function.

(35) The PLL controller **41** is connected to the second voltage sensor **24**, the first converter **42**, and the second converter **43**. The PLL controller **41** acquires information of the grid voltage $v_{sub.Grid}$ as a value detected by the second voltage sensor **24**. The PLL controller **41** performs PLL control based on the grid voltage $v_{sub.Grid}$ and outputs information of a reference phase θ synchronized with the grid voltage $v_{sub.Grid}$ to the first converter **42** and the second converter **43**.

(36) The first converter **42** is connected to the second voltage sensor **24**, the PLL controller **41**, the power controller **44**, and the root mean square value calculator **51**. The first converter **42** acquires the information of the grid voltage $v_{sub.Grid}$ as a value detected by the second voltage sensor **24** and the information of the reference phase θ output from the PLL controller **41**. The first converter **42** performs three-phase two-phase conversion (dq conversion) based on the acquired reference phase θ to convert the grid voltage $v_{sub.Grid}$ into a d-axial voltage value and a q-axial voltage value. Note that the reference phase θ of the dq conversion has a q-axial voltage component of zero, for example. The first converter **42** outputs information of the d-axial voltage value and information of the q-axial voltage value to the power controller **44** and the root mean square value calculator **51**.

(37) The second converter **43** is connected to the second current sensor **23**, the third current sensor **25**, and the power controller **44**. The second converter **43** acquires information of the inverter output current $i_{sub.AC}$ as a value detected by the second current sensor **23**, information of the grid current $i_{sub.Grid}$ as a value detected by the third current sensor **25**, and the information of the reference phase θ output from the PLL controller **41**. The second converter **43** performs three-phase two-phase conversion (dq conversion) based on the acquired reference phase θ to convert the inverter output current $i_{sub.AC}$ and the grid current $i_{sub.Grid}$ into a d-axial current value and a q-axial current value. Note that the reference phase θ of the dq conversion has a q-axial voltage component of zero, for example. The second converter **43** outputs information of the d-axial current value and information of the q-axial current value to the power controller **44**.

(38) The power controller **44** is connected to the first converter **42**, the second converter **43**, the first adder **61**, and the third converter **62**. The power controller **44** acquires the information of the d-axial voltage value and the information of the q-axial voltage value from the first converter **42** and acquires the information of the d-axial current value and the information of the q-axial current value from the second converter **43**. The power controller **44** calculates information of a d-axial current command value $i^*_{sub.d}$ and information of a q-axial current command value $i^*_{sub.q}$ based on the information of the d-axial voltage value, the information of the q-axial voltage value, the information of the d-axial current value, and the information of the q-axial current value thus acquired. The power controller **44** outputs the calculated information of the d-axial current command value $i^*_{sub.d}$ to the first adder **61** and outputs the calculated information of the q-axial

current command value $i^*.sub.q$ to the third converter **62**.

(39) The root mean square value calculator (RMS calculator) **51** is connected to the first converter **42** and the modulation rate calculator **52**. The root mean square value calculator **51** acquires the information of the d-axial voltage value and the information of the q-axial voltage value from the first converter **42**. The root mean square value calculator **51** calculates a grid voltage root mean square value $v.sub.RMS$ based on the information of the d-axial voltage value and the information of the q-axial voltage value thus acquired. The root mean square value calculator **51** outputs information of the calculated grid voltage root mean square value $v.sub.RMS$ to the modulation rate calculator **52**. Note that, in the present specification below, the root mean square value calculator **51** is also referred to as the “RMS calculator **51**”.

(40) The modulation rate calculator **52** is connected to the first voltage sensor **22** and the root mean square value calculator **51**. The modulation rate calculator **52** acquires information of the DC voltage $v.sub.DC$ that is a value detected by the first voltage sensor **22** and the information of the grid voltage root mean square value $v.sub.RMS$ from the root mean square value calculator **51**. The modulation rate calculator **52** calculates a modulation rate based on the information of the DC voltage $v.sub.DC$ and the information of the grid voltage root mean square value $v.sub.RMS$ thus acquired. Note that the modulation rate calculator **52** may calculate the modulation rate constantly or at each predetermined interval or may calculate the modulation rate at optional timings.

(41) The modulation rate calculator **52** includes the limiter adjuster **53** and outputs, for example, the calculated modulation rate to the limiter adjuster **53**. Note that the modulation rate calculator **52** may output information of whether the calculated modulation rate exceeds one (modulation rate $>100\%$) to the limiter adjuster **53**. Alternatively, only when the calculated modulation rate exceeds one (modulation rate $>100\%$), the modulation rate calculator **52** may output the calculated modulation rate or information that the calculated modulation rate exceeds one (modulation rate $>100\%$) to the limiter adjuster **53**.

(42) The limiter adjuster **53** is provided as, for example, part of the modulation rate calculator **52** and is connected to the root mean square value calculator **51** through the modulation rate calculator **52** and also connected to the limiter **54** through the MPPT controller **55**. The limiter adjuster **53** is also connected to the storage **70** although wires are omitted in the drawing. The limiter adjuster **53** acquires information of the modulation rate from the modulation rate calculator **52** and acquires information of a lower bound limiter set value $v.sub.L1$ and information of a rated value $v.sub.R$ from the storage **70**.

(43) The limiter adjuster **53** calculates a lower bound limiter operation value $v.sub.L2$ based on the lower bound limiter set value $v.sub.L1$, the rated value $v.sub.R$, and the grid voltage root mean square value $v.sub.RMS$ thus acquired. Note that the limiter adjuster **53** constantly performs the calculation and updates (calculates) the lower bound limiter operation value $v.sub.L2$ whether the modulation rate exceeds one (modulation rate $>100\%$) or does not exceed one (modulation rate $\leq 100\%$). Then, the limiter adjuster **53** outputs the calculated lower bound limiter operation value $v.sub.L2$ to the limiter **54**. As a result, the modulation rate eventually becomes smaller than one.

(44) Note that the limiter adjuster **53** does not necessarily need to output the lower bound limiter operation value $v.sub.L2$ to the limiter **54** when the lower bound limiter operation value $v.sub.L2$ has no variation as a result of the calculation. Alternatively, the limiter adjuster **53** may output the lower bound limiter operation value $v.sub.L2$ to the limiter **54** constantly, at each predetermined interval, or at optional timings irrespective of the existence of variation in the modulation rate and the existence of variation in the lower bound limiter operation value $v.sub.L2$.

(45) Note that the modulation rate calculator **52** and the limiter adjuster **53** are not limited to the configuration illustrated in FIG. 1, but the modulation rate calculator **52** and the limiter adjuster **53** only need to operate out of synchronization with the MPPT controller **55** at least. Specifically, as long as the modulation rate calculator **52** and the limiter adjuster **53** operate out of synchronization with the MPPT controller **55**, the modulation rate calculator **52** and the limiter adjuster **53** may be

an integrated component or completely separate components, or the limiter adjuster 53 may include the modulation rate calculator 52.

(46) The limiter 54 is provided as, for example, part of the MPPT controller 55 and connected to the limiter adjuster 53 through the modulation rate calculator 52. The limiter 54 acquires information of the lower bound limiter operation value $v_{sub.L2}$ from the limiter adjuster 53. Note that the limiter 54 may acquire the information of the lower bound limiter operation value $v_{sub.L2}$ from the limiter adjuster 53 only when the lower bound limiter operation value $v_{sub.L2}$ has varied. Alternatively, the limiter 54 may acquire the information of the lower bound limiter operation value $v_{sub.L2}$ from the limiter adjuster 53 constantly, at each predetermined interval, or at optional timings.

(47) When having acquired the information of the lower bound limiter operation value $v_{sub.L2}$ from the limiter adjuster 53, the limiter 54 updates the value of a DC voltage command lower bound limiter $v_{sub.L}$ set to the limiter 54. Accordingly, the value of the DC voltage command lower bound limiter $v_{sub.L}$ is updated in accordance with the modulation rate. Note that the limiter 54 does not necessarily need to update the value of the DC voltage command lower bound limiter $v_{sub.L}$ when the lower bound limiter operation value $v_{sub.L2}$ has no variation. The limiter 54 sets the updated lower bound limiter operation value $v_{sub.L2}$ as the present value of the DC voltage command lower bound limiter $v_{sub.L}$.

(48) The maximum power point tracking controller (MPPT controller) 55 is connected to the first current sensor 21, the first voltage sensor 22, and the first subtractor 56. The MPPT controller 55 acquires information of the DC current $i_{sub.DC}$ that is a value detected by the first current sensor 21 and the information of the DC voltage $v_{sub.DC}$ that is a value detected by the first voltage sensor 22. The MPPT controller 55 performs MPPT control based on, for example, a known hill climbing method based on the information of the DC current $i_{sub.DC}$ and the information of the DC voltage $v_{sub.DC}$ thus acquired and calculates a DC voltage command value $v'_{sub.DC}$. The range (limiter range) of the MPPT control by the MPPT controller 55 is determined, for example, for each device. Note that the DC voltage command value $v'_{sub.DC}$ is an example of “first DC voltage command value”.

(49) The MPPT controller 55 includes the limiter 54 and compares the calculated DC voltage command value $v'_{sub.DC}$ and the present DC voltage command lower bound limiter $v_{sub.L}$ set to the limiter 54. As a result of the comparison, when the calculated DC voltage command value $v'_{sub.DC}$ is smaller than the DC voltage command lower bound limiter $v_{sub.L}$ set to the limiter 54, the MPPT controller 55 determines the value of the DC voltage command lower bound limiter $v_{sub.L}$ to be the next DC voltage command value $v_{sub.DC}$. When the calculated DC voltage command value $v'_{sub.DC}$ is equal to or larger than the value of the DC voltage command lower bound limiter $v_{sub.L}$ set to the limiter 54, the MPPT controller 55 determines the calculated DC voltage command value $v'_{sub.DC}$ to be the next DC voltage command value $v_{sub.DC}$. The MPPT controller 55 outputs information of the determined DC voltage command value $v_{sub.DC}$ to the first subtractor 56. Note that the DC voltage command value $v_{sub.DC}$ is an example of “second DC voltage command value”.

(50) The first subtractor 56 is connected to the first voltage sensor 22 and the MPPT controller 55. The first subtractor 56 acquires the information of the DC voltage $v_{sub.DC}$ that is a value detected by the first voltage sensor 22 and the information of the DC voltage command value $v_{sub.DC}$ output from the MPPT controller 55. The first subtractor 56 calculates a value by subtracting the DC voltage $v_{sub.DC}$ from the DC voltage command value $v_{sub.DC}$. The first subtractor 56 outputs information of the value calculated by subtracting the DC voltage $v_{sub.DC}$ from the DC voltage command value $v_{sub.DC}$ to the DC voltage controller 57.

(51) The DC voltage controller 57 is connected to the first subtractor 56 and the first adder 61. The DC voltage controller 57 acquires the information of the value calculated by subtracting the DC voltage $v_{sub.DC}$ from the DC voltage command value $v_{sub.DC}$ and output from the first

subtractor **56**. The DC voltage controller **57** calculates a d-axial current command value by controlling the acquired information. The DC voltage controller **57** outputs information of the calculated d-axial current command value to the first adder **61**.

(52) The first adder **61** is connected to the power controller **44**, the DC voltage controller **57**, and the third converter **62**. The first adder **61** acquires the information of the d-axial current command value $i^*.sub.d$ output from the power controller **44** and the information of the d-axial current command value output from the DC voltage controller **57**. The first adder **61** calculates a value by adding the d-axial current command value $i^*.sub.d$ and the d-axial current command value thus acquired. The first adder **61** outputs information of the value calculated by adding the d-axial current command value $i^*.sub.d$ and the d-axial current command value to the third converter **62**.

(53) The third converter **62** is connected to the power controller **44**, the first adder **61**, and the second subtractor **63**. The third converter **62** acquires the information of the q-axial current command value $i^*.sub.q$ output from the power controller **44** and the information of the value obtained by adding the d-axial current command value $i^*.sub.d$ and the d-axial current command value and output from the first adder **61**. The third converter **62** calculates a current command value $i^*.sub.AC$ through two-phase three-phase conversion (inverse dq conversion) of the acquired information. The third converter **62** outputs information of the calculated current command value $i^*.sub.AC$ to the second subtractor **63**.

(54) The second subtractor **63** is connected to the second current sensor **23**, the third converter **62**, and the current controller **64**. The second subtractor **63** acquires the information of the inverter output current $i.sub.AC$ as a value detected by the second current sensor **23** and the information of the current command value $i^*.sub.AC$ output from the third converter **62**. The second subtractor **63** calculates a value by subtracting the inverter output current $i.sub.AC$ from the current command value $i^*.sub.AC$. The second subtractor **63** outputs information of the value calculated by subtracting the inverter output current $i.sub.AC$ from the current command value $i^*.sub.AC$ to the current controller **64**.

(55) The current controller **64** is connected to the second subtractor **63** and the PWM controller **65**. The current controller **64** acquires the information of the value calculated by subtracting the inverter output current $i.sub.AC$ from the current command value $i^*.sub.AC$ and output from the second subtractor **63**. The current controller **64** calculates a voltage command value by controlling the acquired information. The current controller **64** outputs the calculated voltage command value to the PWM controller **65**.

(56) The pulse width modulation controller (PWM controller) **65** is connected to the inverter **13** and the current controller **64**. The PWM controller **65** acquires the voltage command value from the current controller **64**. The PWM controller **65** performs PWM control based on the acquired voltage command value to generate a gate signal as the pulse width modulation signal (PWM signal). The PWM controller **65** outputs the generated gate signal to the inverter **13** and collectively controls operation of the inverter **13** by controlling the non-illustrated switching elements of the inverter **13**. Note that the PWM controller **65** is an example of “inverter controller”.

(57) The storage **70** is a volatile or non-volatile storage medium such as a hard disk drive (HDD), a solid state drive (SSD), a dynamic random access memory (DRAM), or any other semiconductor memory. The storage **70** is connected to each component of the control device **30** such that various kinds of information can be input and output through, for example, a non-illustrated bus (system bus). The storage **70** stores, for example, a computer program necessary for operation of each component of the control device **30** and is subjected to writing and reading of various kinds of information by each component of the control device **30**.

(58) The storage **70** stores a value acquired by each sensor such as the first current sensor **21**. The storage **70** also stores, for example, various calculation formulae, coefficients, thresholds, and the like used for calculation by components such as the limiter adjuster **53** and the MPPT controller **55**, predetermined rated values, and limiter values. The storage **70** also stores, for example, results of

calculation by components such as the root mean square value calculator **51**, the modulation rate calculator **52**, the limiter adjuster **53**, and the MPPT controller **55**.

(59) Note that the storage **70** may be provided outside the control device **30** and connected to the control device **30** in a wired or wireless manner and may be an external storage medium such as a memory card or a digital versatile disc (DVD) or may be an online storage or the like. The storage **70** may be the same as the memory **92** (refer to FIG. **9**) to be described later.

(60) FIG. **2** is a diagram illustrating an example of the configuration and processing of the modulation rate calculator **52** and the limiter adjuster **53** illustrated in FIG. **1**.

(61) Although wires are omitted in the drawing, the modulation rate calculator **52** is connected to the first voltage sensor **22** and the root mean square value calculator **51** as described above with reference to FIG. **1**, acquires the information of the DC voltage $v_{\text{sub.DC}}$ from the first voltage sensor **22**, and acquires the information of the grid voltage root mean square value $v_{\text{sub.RMS}}$ from the root mean square value calculator **51**. The modulation rate calculator **52** calculates the modulation rate based on the information of the DC voltage $v_{\text{sub.DC}}$ and the information of the grid voltage root mean square value $v_{\text{sub.RMS}}$ thus acquired.

(62) The modulation rate calculator **52** calculates the modulation rate by, for example, Expression (1) below as exemplarily illustrated at a lower part of FIG. **2**. Note that, in Expression (1), “grid voltage root mean square value” $\times\sqrt{2}$ is a peak value of the grid voltage.

(63) “modulationrate” = “gridvoltage root meansquarevalue” $\times\sqrt{2}$ / “DCvoltage” (1)

(64) Note that, as illustrated at the lower part of FIG. **2**, occurrence of overmodulation is determined when the modulation rate is larger than 100% (modulation rate $>100\%$), in other words, when the modulation rate exceeds one.

(65) A case in which the grid voltage root mean square value $v_{\text{sub.RMS}}$ is 600 V and the DC voltage $v_{\text{sub.DC}}$ is 1000 V will be described below as an example. In this case, the modulation rate is calculated as $600\text{ V}\times\sqrt{2}/1000\text{ V}=84.9\%$ based on Expression (1). In this case, overmodulation does not occur since the modulation rate does not exceed one (modulation rate $\leq 100\%$). Note that the power conversion system **1** can stably operate when overmodulation does not occur.

(66) A case in which the grid voltage root mean square value $v_{\text{sub.RMS}}$ increases by 10% to 660 V and the DC voltage $v_{\text{sub.DC}}$ simultaneously decreases to 930 V will be described below as another example. In this case, the modulation rate is calculated as $660\text{ V}\times\sqrt{2}/930\text{ V}=100.4\%$ based on Expression (1). In this case, overmodulation occurs since the modulation rate exceeds one (modulation rate $>100\%$). Note that the power conversion system **1** cannot stably operate when overmodulation occurs.

(67) As described above with reference to FIG. **1**, the modulation rate calculator **52** outputs the calculated modulation rate or information of whether the calculated modulation rate exceeds one (modulation rate $>100\%$) to the limiter adjuster **53**. Alternatively, only when the calculated modulation rate exceeds one (modulation rate $>100\%$), the modulation rate calculator **52** may output the calculated modulation rate or information that the calculated modulation rate exceeds one (modulation rate $>100\%$) to the limiter adjuster **53**.

(68) Although wires are omitted in the drawing, the limiter adjuster **53** is connected to the root mean square value calculator **51**, the limiter **54**, and the storage **70** as described above with reference to FIG. **1**. The limiter adjuster **53** includes, for example, a lower bound limiter set value acquirer **53a**, a grid voltage root mean square value acquirer **53b**, a rated value acquirer **53c**, a divider **53d**, an upper-lower bound limiter **53e**, a multiplier **53f**, and a lower bound limiter operation value outputter **53g**.

(69) The lower bound limiter set value acquirer **53a** is connected to the storage **70** and the multiplier **53f**. The lower bound limiter set value acquirer **53a** acquires the lower bound limiter set value $v_{\text{sub.L1}}$ of MPPT reference voltage from the storage **70** and outputs the acquired lower

bound limiter set value $v_{sub.L1}$ to the multiplier **53f**. Note that the lower bound limiter set value $v_{sub.L1}$ of MPPT reference voltage is determined to be a predetermined value for each device as a specification, for example. In the example illustrated in FIG. 2, the lower bound limiter set value acquirer **53a** acquires 900 V as the lower bound limiter set value $v_{sub.L1}$ from the storage **70** and outputs 900 V thus acquired to the multiplier **53f**.

(70) The grid voltage root mean square value acquirer **53b** is connected to the root mean square value calculator **51** and the divider **53d**. The grid voltage root mean square value acquirer **53b** acquires the grid voltage root mean square value $v_{sub.RMS}$ calculated by the root mean square value calculator **51** from the root mean square value calculator **51** and outputs the acquired grid voltage root mean square value $v_{sub.RMS}$ to the divider **53d**. In the example illustrated in FIG. 2, the grid voltage root mean square value acquirer **53b** acquires 630 V as the grid voltage root mean square value $v_{sub.RMS}$ from the root mean square value calculator **51** and outputs 630 V thus acquired to the divider **53d**.

(71) The rated value acquirer **53c** is connected to the storage **70** and the divider **53d**. The rated value acquirer **53c** acquires the rated value $v_{sub.R}$ from the storage **70** and outputs the acquired rated value $v_{sub.R}$ to the divider **53d**. Note that the rated value $v_{sub.R}$ is determined to be a predetermined value for each device as a specification, for example. In the example illustrated in FIG. 2, the rated value acquirer **53c** acquires 600 V as the rated value $v_{sub.R}$ from the storage **70** and outputs 600 V thus acquired to the divider **53d**.

(72) The divider **53d** is connected to the grid voltage root mean square value acquirer **53b**, the rated value acquirer **53c**, and the upper-lower bound limiter **53e**. The divider **53d** acquires the grid voltage root mean square value $v_{sub.RMS}$ from the grid voltage root mean square value acquirer **53b**, acquires the rated value $v_{sub.R}$ from the rated value acquirer **53c**, and divides the acquired grid voltage root mean square value $v_{sub.RMS}$ by the rated value $v_{sub.R}$. Then, the divider **53d** outputs a value obtained by dividing the grid voltage root mean square value $v_{sub.RMS}$ by the rated value $v_{sub.R}$ to the upper-lower bound limiter **53e**.

(73) In the example illustrated in FIG. 2, the divider **53d** acquires 630 V as the grid voltage root mean square value $v_{sub.RMS}$ from the grid voltage root mean square value acquirer **53b**, acquires 600 V as the rated value $v_{sub.R}$ from the rated value acquirer **53c**, and divides 630 V thus acquired by 600 V. Then, the divider **53d** outputs 1.05 that is a value obtained by dividing 630 V by 600 V to the upper-lower bound limiter **53e**.

(74) The upper-lower bound limiter **53e** is connected to the divider **53d** and the multiplier **53f**. The upper-lower bound limiter **53e** acquires the value obtained by dividing the grid voltage root mean square value $v_{sub.RMS}$ by the rated value $v_{sub.R}$ from the divider **53d**. Then, the upper-lower bound limiter **53e** compares the acquired value with an upper bound value and a lower bound value (upper and lower bound values) set to the upper-lower bound limiter **53e**. When the acquired value is in the range between the upper and lower bound values, the upper-lower bound limiter **53e** outputs the acquired value as a value limited by the upper-lower bound limiter **53e** to the multiplier **53f**. When the acquired value is out of the range between the upper and lower bound values, the upper-lower bound limiter **53e** outputs, as the value limited by the upper-lower bound limiter **53e** to the multiplier **53f**, a value obtained by limiting the acquired value to inside of the range between the upper and lower bound values.

(75) In the example illustrated in FIG. 2, the upper-lower bound limiter **53e** acquires 1.05 that is the value obtained by dividing 630 V by 600 V from the divider **53d**. Then, since 1.05 thus acquired is in the range between the upper bound value 1.1 and the lower bound value 1.0, the upper-lower bound limiter **53e** outputs 1.05 thus acquired to the multiplier **53f** as the value limited by the upper-lower bound limiter **53e**.

(76) Note that the upper and lower bound values set to the upper-lower bound limiter **53e** may be determined to be predetermined values for each device as a specification, for example, and the lower bound value may be 1.0. Accordingly, the lower bound limiter operation value $v_{sub.L2}$ is

controlled only when the grid voltage root mean square value $v_{sub.RMS}$ is larger than the rated value $v_{sub.R}$, which can prevent the lower bound limiter operation value $v_{sub.L2}$ from becoming smaller than the lower bound limiter set value $v_{sub.L1}$ of MPPT reference voltage.

(77) The multiplier **53f** is connected to the lower bound limiter set value acquirer **53a**, the upper-lower bound limiter **53e**, and the lower bound limiter operation value outputter **53g**. The multiplier **53f** acquires the lower bound limiter set value $v_{sub.L1}$ from the lower bound limiter set value acquirer **53a** and acquires the value limited by the upper-lower bound limiter **53e** from the upper-lower bound limiter **53e**. The multiplier **53f** multiplies the acquired lower bound limiter set value $v_{sub.L1}$ by the acquired value limited by the upper-lower bound limiter **53e**. The multiplier **53f** outputs a value obtained by multiplying the lower bound limiter set value $v_{sub.L1}$ by the value limited by the upper-lower bound limiter **53e** to the lower bound limiter operation value outputter **53g**.

(78) In the example illustrated in FIG. 2, the multiplier **53f** acquires 900 V as the lower bound limiter set value $v_{sub.L1}$ from the lower bound limiter set value acquirer **53a** and acquires 1.05 as the value limited by the upper-lower bound limiter **53e** from the upper-lower bound limiter **53e**. Then, the multiplier **53f** outputs 945 V that is a value obtained by multiplying 900 V by 1.05 to the lower bound limiter operation value outputter **53g**.

(79) The lower bound limiter operation value outputter **53g** is connected to the multiplier **53f** and the limiter **54**. The lower bound limiter operation value outputter **53g** acquires the value obtained by multiplying the lower bound limiter set value $v_{sub.L1}$ by the value limited by the upper-lower bound limiter **53e** from the multiplier **53f** and outputs the acquired value as the lower bound limiter operation value $v_{sub.L2}$ to the limiter **54**. In the example illustrated in FIG. 2, the lower bound limiter operation value outputter **53g** acquires 945 V from the multiplier **53f** and outputs 945 V thus acquired to the limiter **54** as the lower bound limiter operation value $v_{sub.L2}$.

(80) Through the above-described processing, when having acquired the information of the lower bound limiter operation value $v_{sub.L2}$ from the limiter adjuster **53** (lower bound limiter operation value outputter **53g**), the limiter **54** (refer to FIG. 1) updates the value of the DC voltage command lower bound limiter $v_{sub.L}$ set to the limiter **54**. Accordingly, it is possible to update the value of the DC voltage command lower bound limiter $v_{sub.L}$ in accordance with the modulation rate, thereby achieving a wider operation range on the DC side than in conventional cases and stably operating the power conversion system **1** without overmodulation.

(81) Specifically, the DC voltage command lower bound limiter $v_{sub.L}$ needs to be set with consideration on the operation range of the grid voltage of the power conversion system **1**, but the upper bound of the grid voltage typically needs to be up to 110% of rated voltage due to cooperative regulations among countries. However, the grid voltage is near 100% of rated voltage most of every day. Thus, for example, when the DC voltage command lower bound limiter $v_{sub.L}$ is set to be higher than a device specification in advance to avoid overmodulation based on assumption of 110% of rated voltage, the operation range on the DC side becomes narrow. In the example illustrated in FIG. 2, the limiter range of MPPT voltage is 900 V to 1300 V as illustrated at the lower part of FIG. 2, but when the lower bound voltage is increased with consideration on increase of the grid voltage, the operation range becomes, for example, 990 V to 1300 V and accordingly, the operation range on the DC side becomes narrow.

(82) However, according to the present embodiment, for example, it is possible to, in accordance with the modulation rate, increase the DC voltage command lower bound limiter $v_{sub.L}$ when the grid voltage root mean square value $v_{sub.RMS}$ has increased, and decrease the DC voltage command lower bound limiter $v_{sub.L}$ when the grid voltage root mean square value $v_{sub.RMS}$ has decreased. Accordingly, it is possible to update the value of the DC voltage command lower bound limiter $v_{sub.L}$ in accordance with the modulation rate, thereby achieving a wider operation range on the DC side than in conventional cases and stably operating the power conversion system **1** without overmodulation.

Operation of First Embodiment

(83) FIG. 3 is a flowchart illustrating an example of operation of the MPPT controller 55 illustrated in FIG. 1. The flowchart illustrated in FIG. 3 is started when operation of the power conversion system 1 is started.

(84) At step S1, the MPPT controller 55 calculates a DC voltage command value 1 changed to a value obtained by adding or subtracting ΔV to or from the present DC voltage command value $v^*.sub.DC$. For example, the MPPT controller 55 calculates the DC voltage command value 1 changed to the value obtained by adding ΔV to the present DC voltage command value $v^*.sub.DC$. Note that the addition and subtract may be performed, for example, in an order in accordance with a rule predetermined for each device or in an optional order.

(85) At step S2, the MPPT controller 55 calculates first power-calculation power $P.sub.1$ when the DC voltage command value $v^*.sub.DC$ is the DC voltage command value 1.

(86) At step S3, oppositely to step S1, the MPPT controller 55 calculates the DC voltage command value 1 changed to the value obtained by adding or subtracting ΔV to or from the present DC voltage command value $v^*.sub.DC$. For example, when having added ΔV to the present DC voltage command value $v^*.sub.DC$ at step S1, the MPPT controller 55 calculates, oppositely to step S1, a DC voltage command value 2 changed to the value obtained by subtracting ΔV from the present DC voltage command value $v^*.sub.DC$. For example, when having subtracted ΔV from the present DC voltage command value $v^*.sub.DC$ at step S1, the MPPT controller 55 calculates, oppositely to step S1, the DC voltage command value 2 changed to the value obtained by adding ΔV to the present DC voltage command value $v^*.sub.DC$. At step S1 described above, the MPPT controller 55 has added ΔV to the present DC voltage command value $v^*.sub.DC$, and thus at step S3, oppositely to step S1, the MPPT controller 55 calculates the DC voltage command value 2 changed to the value obtained by subtracting ΔV from the present DC voltage command value $v^*.sub.DC$.

(87) At step S4, the MPPT controller 55 calculates second power-calculation power $P.sub.2$ when the DC voltage command value $v^*.sub.DC$ is the DC voltage command value 2.

(88) At step S5, the MPPT controller 55 compares the first power-calculation power $P.sub.1$ calculated at step S2 and the second power-calculation power $P.sub.2$ calculated at step S4 and determines whether the first power-calculation power $P.sub.1$ is larger than the second power-calculation power $P.sub.2$. Then, when the first power-calculation power $P.sub.1$ is larger than the second power-calculation power $P.sub.2$ (Yes), the MPPT controller 55 advances the processing to step S6. When the first power-calculation power $P.sub.1$ is not larger than the second power-calculation power $P.sub.2$ (No), the MPPT controller 55 advances the processing to step S7. In other words, when the second power-calculation power $P.sub.2$ is larger than the first power-calculation power $P.sub.1$, the MPPT controller 55 advances the processing to step S7.

(89) At step S6, the MPPT controller 55 performs control to change the next DC voltage command value $v^*.sub.DC$ to the DC voltage command value 1. This is because power is larger when the next DC voltage command value $v^*.sub.DC$ is changed to the DC voltage command value 1 than when the next DC voltage command value $v^*.sub.DC$ is changed to the DC voltage command value 2.

(90) At step S7, the MPPT controller 55 performs control to change the next DC voltage command value $v^*.sub.DC$ to the DC voltage command value 2. This is because power is larger when the next DC voltage command value $v^*.sub.DC$ is changed to the DC voltage command value 2 than when the next DC voltage command value $v^*.sub.DC$ is changed to the DC voltage command value 1.

(91) FIG. 4 is a diagram for description of an example of MPPT control based on a known hill climbing method by the MPPT controller 55 illustrated in FIG. 1. In FIG. 4, the vertical axis represents power P , and the horizontal axis represents voltage V . In FIG. 4, two upward curves are curves representing output characteristics when the solar battery receives light and generates

power, and are called power voltage characteristic curves. Note that, in the present specification below, a power voltage characteristic curve representing output characteristics of the solar battery is also referred to as a “power voltage (PV) curve”.

(92) For example, a PV curve changes due to change of a surrounding condition such as solar irradiation or temperature for the same solar panel (solar battery panel). PV curves PV.sub.A and PV.sub.B are examples of a PV curve that has changed due to change of a surrounding condition. An optimum operation point P.sub.mppA on the PV curve PV.sub.A is a local maximum point at which the product of operation current and operation voltage is largest, and open voltage V.sub.OCA is voltage in a state in which an output terminal of the solar battery is connected to no load nor the like and opened. Similarly, an optimum operation point P.sub.mppB on the PV curve PV.sub.B is a local maximum point at which the product of operation current and operation voltage is largest, and open voltage V.sub.OCB is voltage in a state in which the output terminal of the solar battery is connected to no load nor the like and opened.

(93) For example, the following describes a case in which the present DC voltage command value is the DC voltage command value $v^*.sub.DC$ when the present PV curve is the PV curve PV.sub.A. In this case, the MPPT controller 55 calculates the DC voltage command value 1 obtained by adding ΔV to the present DC voltage command value $v^*.sub.DC$ (S1) and calculates first power-calculation power P1A in this case (S2). Thereafter, the MPPT controller 55 calculates the DC voltage command value 2 obtained by subtracting ΔV from the present DC voltage command value $v^*.sub.DC$ (S3) and calculates second power-calculation power P.sub.2A in this case (S4). Then, the MPPT controller 55 compares the power P.sub.1A and the power P.sub.2A (S5) and performs control to change the next DC voltage command value $v'^*.sub.DC$ to the DC voltage command value 2 since the power P.sub.2A is larger than the power P.sub.1A according to the graph (S7).

(94) For example, when the present PV curve is the PV curve PV.sub.B, the MPPT controller 55 calculates the DC voltage command value 1 obtained by adding ΔV to the present DC voltage command value $v^*.sub.DC$ (S1) and calculates first power-calculation power P.sub.1B in this case (S2). Thereafter, the MPPT controller 55 calculates the DC voltage command value 2 obtained by subtracting ΔV from the present DC voltage command value $v^*.sub.DC$ (S3) and calculates second power-calculation power P.sub.2B in this case (S4). Then, the MPPT controller 55 compares the power P.sub.1B and the power P.sub.2B (S5) and performs control to change the next DC voltage command value $v'^*.sub.DC$ to the DC voltage command value 1 since the power P.sub.1B is larger than the power P.sub.2B according to the graph (S6).

(95) Accordingly, the MPPT controller 55 constantly performs, for example, the above-described MPPT control based on a known hill climbing method and thus can operate the power conversion system 1 at an optimum operation point P.sub.mpp at which power is constantly maximum even when the PV curve has changed due to change of a surrounding condition. In other words, the MPPT controller 55 constantly performs, for example, the above-described MPPT control based on a known hill climbing method and thus can operate the power conversion system 1 at the optimum operation point P.sub.mpp at which power is constantly maximum even when the position of the apex (optimum operation point P.sub.mpp) of the PV curve has changed.

(96) As illustrated in FIG. 3, when having acquired the information of the lower bound limiter operation value $v.sub.L2$ from the limiter adjuster 53 at step S8, the limiter 54 of the MPPT controller 55 updates the value of the DC voltage command lower bound limiter $v.sub.L$ set to the limiter 54. Note that the limiter 54 does not necessarily need to update the value of the DC voltage command lower bound limiter $v.sub.L$ when the lower bound limiter operation value $v.sub.L2$ has no variation. The limiter 54 sets the updated lower bound limiter operation value $v.sub.L2$ as the present value of the DC voltage command lower bound limiter $v.sub.L$. Note that the operation at step S8 is performed out of synchronization with the MPPT control performed by the MPPT controller 55 as described above with reference to, for example, FIG. 2. Thus, the operation at step S8 does not necessarily need to be performed at the timing indicated at step S8 but is performed at

an optional timing.

(97) At step S9, the MPPT controller 55 determines whether the DC voltage command value $v'^{*}.sub.DC$ determined at step S6 or S7 is smaller than the present DC voltage command lower bound limiter $v.sub.L$ set to the limiter 54. When the DC voltage command value $v'^{*}.sub.DC$ is smaller than the DC voltage command lower bound limiter $v.sub.L$ set to the limiter 54 (Yes), the MPPT controller 55 advances the processing to step S10. When the DC voltage command value $v'^{*}.sub.DC$ is equal to or larger than the value of the DC voltage command lower bound limiter $v.sub.L$ set to the limiter 54 (No), the MPPT controller 55 advances the processing to step S11.

(98) At step S10, the MPPT controller 55 determines the value of the DC voltage command lower bound limiter $v.sub.L$ to be the next DC voltage command value $v^{*}.sub.DC$. Accordingly, overmodulation can be prevented even when the grid voltage $v.sub.Grid$ (grid voltage root mean square value $v.sub.RMS$) varies.

(99) At step S11, the MPPT controller 55 determines the DC voltage command value $v'^{*}.sub.DC$ as the next DC voltage command value $v^{*}.sub.DC$. This is because no overmodulation occurs when the DC voltage command value $v^{*}.sub.DC$ is larger than the DC voltage command lower bound limiter $v.sub.L$.

(100) At step S12, the MPPT controller 55 outputs the DC voltage command value $v^{*}.sub.DC$ determined at step S10 or S11 to the first subtractor 56 (refer to FIG. 1).

(101) At step S13, the MPPT controller 55 determines whether the MPPT control has ended. When having determined that the MPPT control has ended (Yes), the MPPT controller 55 ends the processing of the flowchart of FIG. 3. When having determined that the MPPT control has not ended (No), the MPPT controller 55 returns the processing to step S1 and repeats the processing at steps S1 to S13.

(102) FIG. 5 is a diagram for description of an example of specific processing in the operation of the MPPT controller 55 described above with reference to FIG. 3. In FIG. 5, the vertical axis represents voltage V and the horizontal axis represents time point t , and time flows from left to right. Straight line a represents the DC voltage command value $v^{*}.sub.DC$ or $v'^{*}.sub.DC$, thin straight line b represents the DC voltage command value 1 or 2, and thick dashed line c represents the DC voltage command lower bound limiter $v.sub.L$. Dashed line a' represents the DC voltage command value $v^{*}.sub.DC$ or $v'^{*}.sub.DC$ when it is assumed that the DC voltage command lower bound limiter $v.sub.L$ is not updated. Thin dashed line b' represents the DC voltage command value 1 or 2 when it is assumed that the DC voltage command lower bound limiter $v.sub.L$ is not updated. Dashed line c' represents the DC voltage command lower bound limiter $v.sub.L$ when it is assumed that the DC voltage command lower bound limiter $v.sub.L$ is not updated.

(103) At time point $t.sub.1$, the MPPT controller 55 calculates the DC voltage command value 1 that is a value obtained by adding ΔV to the DC voltage command value $v^{*}.sub.DC$.

(104) At time point $t.sub.2$, the MPPT controller 55 calculates the first power-calculation power $P.sub.1$ when the DC voltage command value $v^{*}.sub.DC$ is the DC voltage command value 1.

(105) At time point $t.sub.3$, the MPPT controller 55 calculates the DC voltage command value 2 that is a value obtained by subtracting ΔV from the DC voltage command value $V^{*}.sub.DC$.

(106) At time point $t.sub.4$, the MPPT controller 55 calculates the second power-calculation power $P.sub.2$ when the DC voltage command value $v^{*}.sub.DC$ is the DC voltage command value 2.

(107) At time point $t.sub.5$ (MPPT control period), the MPPT controller 55 compares the power $P.sub.1$ and the power $P.sub.2$ and determines that the power $P.sub.2$ is larger than the power $P.sub.1$, and thus performs control to change the next DC voltage command value $v'^{*}.sub.DC$ to the DC voltage command value 2. Since the changed DC voltage command value $v'^{*}.sub.DC$ is larger than the DC voltage command lower bound limiter $v.sub.L$, the MPPT controller 55 determines the DC voltage command value $v'^{*}.sub.DC$ to be the next DC voltage command value $v^{*}.sub.DC$. The MPPT controller 55 outputs the determined DC voltage command value $v^{*}.sub.DC$ to the first subtractor 56 (refer to FIG. 1). Simultaneously, at time point $t.sub.5$, the MPPT controller 55

calculates the DC voltage command value **1** that is a value obtained by subtracting ΔV from the DC voltage command value $v^*.sub.DC$.

(108) At time point $t.sub.6$, the MPPT controller **55** calculates the first power-calculation power $P.sub.1$ when the DC voltage command value $v^*.sub.DC$ is the DC voltage command value **1**.

(109) At time point $t.sub.7$, the MPPT controller **55** calculates the DC voltage command value **2** that is a value obtained by adding ΔV to the DC voltage command value $v^*.sub.DC$.

(110) At time point $t.sub.8$ (at a timing out of synchronization with the MPPT control, by the limiter adjuster **53**), the value of the DC voltage command lower bound limiter $v.sub.L$ set to the limiter **54** is updated. For example, when the grid voltage root mean square value $v.sub.RMS$ is 105%, the value of the DC voltage command lower bound limiter $v.sub.L$ set to the limiter **54** is updated to increase by 5%.

(111) At time point $t.sub.9$, the MPPT controller **55** calculates the second power-calculation power $P.sub.2$ when the DC voltage command value $v^*.sub.DC$ is the DC voltage command value **2**.

(112) At time point $t.sub.10$ (MPPT control period), the MPPT controller **55** compares the power $P.sub.1$ and the power $P.sub.2$ and determines that the power $P.sub.1$ is larger than the power $P.sub.2$, and thus performs control to change the next DC voltage command value $v'^*.pc$ to the DC voltage command value **1** (dashed line a'). However, since the changed DC voltage command value $v'^*.sub.DC$ (dashed line a') is smaller than the DC voltage command lower bound limiter $v.sub.L$ (thick dashed line c), the MPPT controller **55** determines the value of the DC voltage command lower bound limiter $v.sub.L$ to be the next DC voltage command value $v^*.sub.DC$. The MPPT controller **55** outputs the determined DC voltage command value $v^*.sub.DC$ to the first subtractor **56** (refer to FIG. 1). Simultaneously at time point $t.sub.10$, the MPPT controller **55** calculates the DC voltage command value **1** that is a value obtained by subtracting ΔV from the DC voltage command value $v^*.sub.DC$ (straight line a) having increased to the value of the DC voltage command lower bound limiter $v.sub.L$.

(113) At time point $t.sub.11$, the MPPT controller **55** calculates the first power-calculation power $P.sub.1$ when the DC voltage command value $v^*.sub.DC$ is the DC voltage command value **1**.

(114) At time point $t.sub.12$, the MPPT controller **55** calculates the DC voltage command value **2** that is a value obtained by adding ΔV to the DC voltage command value $v^*.sub.DC$ (straight line a) having increased to the value of the DC voltage command lower bound limiter $v.sub.L$.

(115) At time point $t.sub.13$, the MPPT controller **55** calculates the second power-calculation power $P.sub.2$ when the DC voltage command value $v^*.sub.DC$ is the DC voltage command value **2**.

(116) At time point $t.sub.14$ (MPPT control period) and later, the MPPT controller **55** repeats the above-described processing.

(117) Note that, when the DC voltage command lower bound limiter $v.sub.L$ is not updated, the MPPT controller **55** determines that the changed DC voltage command value $v'^*.sub.DC$ (dashed line a') is larger than the DC voltage command lower bound limiter $v.sub.L$ (dashed line c') at time point $t.sub.10$ (MPPT control period). In this case, the MPPT controller **55** determines the DC voltage command value $v'^*.sub.DC$ (dashed line a') to be the next DC voltage command value $v^*.sub.DC$.

(118) In this case, overmodulation potentially occurs and the power conversion system **1** potentially cannot stably operate, for example, when the grid voltage root mean square value $v.sub.RMS$ is 105%. Furthermore, the operation range on the DC side is narrow when the DC voltage command lower bound limiter $v.sub.L$ is set to be high in advance to avoid overmodulation.

(119) However, according to the present embodiment, since the DC voltage command lower bound limiter $v.sub.L$ is updated in accordance with the modulation rate, it is possible to stably operate the power conversion system **1** without overmodulation while not narrowing the operation range on the DC side.

Effects of First Embodiment

(120) As described above, according to the first embodiment illustrated in FIGS. 1 to 5, it is

possible to update the DC voltage command lower bound limiter $v_{sub.L}$ in accordance with the modulation rate, thereby achieving a wider operation range on the DC side than in conventional cases and stably operating the power conversion system **1** without overmodulation.

(121) In addition, according to the first embodiment illustrated in FIGS. **1** to **5**, the update of the DC voltage command lower bound limiter $v_{sub.L}$ is performed by the modulation rate calculator **52** and the limiter adjuster **53**, separately from the MPPT controller **55**. In other words, the control to update the DC voltage command lower bound limiter $v_{sub.L}$ and the MPPT control are independently performed in parallel by separate components. Thus, the MPPT controller **55** only needs to determine whether the DC voltage command value $v^*_{sub.DC}$ is smaller than the DC voltage command lower bound limiter $v_{sub.L}$ set to the limiter **54**, which simplifies the control by the MPPT controller **55**. Moreover, the MPPT control by the MPPT controller **55** is not affected by the control to update the DC voltage command lower bound limiter $v_{sub.L}$. Furthermore, since the control to update the DC voltage command lower bound limiter $v_{sub.L}$ and the MPPT control are performed out of synchronization, the control to update the DC voltage command lower bound limiter $v_{sub.L}$ can be swiftly performed at a correct timing in accordance with the modulation rate.

Configuration of Second Embodiment

(122) FIG. **6** is a diagram illustrating an example of the configuration of a control device **30A** and a power conversion system **1A** according to a second embodiment. Note that, in FIG. **6**, any component identical or similar to a component illustrated in FIG. **1** is denoted by the same reference sign, and detailed description thereof is omitted or simplified. In FIG. **6**, the power conversion system **1A** is connected to an energy storage system **4A** through the DC bus **2** on the one end side (input side) on the left side in FIG. **1**.

(123) For example, the power conversion system (PCS) **1A** converts DC power supplied from the energy storage system **4A** into AC power and outputs the converted AC power to the AC power grid **5** side (the AC side). In other words, the power conversion system **1A** in the present embodiment is an energy-storage-system (ESS) power conversion system (ESS-PCS: Energy Storage System-Power Conditioning System). Hereinafter, in the present specification, the power conversion system **1A** is also referred to as the “ESS-PCS **1A**” or simply as the “PCS **1A**”.

(124) The energy storage system **4A** is connected to the DC end (input end) of the inverter **13** through the DC bus **2**. The energy storage system **4A** is also referred to as, for example, a secondary battery or a battery and is a chemical battery that can be repeatedly used through charging and discharging and can supply stored power as necessary. The energy storage system **4A** may be, for example, a lead battery, a nickel-cadmium battery, a nickel-hydrogen battery, or a lithium-ion battery. The energy storage system **4A** supplies stored DC power to the inverter **13** through the DC bus **2**. Note that, in the present specification below, the energy storage system **4A** is also referred to as the “ESS **4A**”.

(125) As illustrated in FIG. **6**, the control device **30A** includes components or functions as a modulation rate calculator **82**, a limiter adjuster **83**, a limiter **84**, a DC voltage operation range detector **85**, an operation controller **86**, and a failure detector **87**. Note that each above-described component or function may be implemented by a computer program executed by the processor **91** (refer to FIG. **9**) to be described later in the processing circuit **90** (refer to FIG. **9**) included in the control device **30A** and to be described later or may be implemented by the hardware component **93** (refer to FIG. **9**) to be described later. The control device **30A** executes a predetermined computer program and performs processing below with each above-described component or function.

(126) The modulation rate calculator **82** is connected to the first voltage sensor **22** and the root mean square value calculator **51**. The modulation rate calculator **82** acquires information of the DC voltage $v_{sub.DC}$ that is a value detected by the first voltage sensor **22** and information of the grid voltage root mean square value $v_{sub.RMS}$ from the root mean square value calculator **51**. The modulation rate calculator **82** calculates the modulation rate based on the information of the DC

voltage v.sub.DC and the information of the grid voltage root mean square value v.sub.RMS thus acquired. Note that the modulation rate calculator **82** may calculate the modulation rate constantly or at each predetermined interval or may calculate the modulation rate at optional timings.

(127) The modulation rate calculator **82** includes the limiter adjuster **83** and outputs, for example, the calculated modulation rate to the limiter adjuster **83**. Note that the modulation rate calculator **82** may output information of whether the calculated modulation rate exceeds one (modulation rate >100%) to the limiter adjuster **83**. Alternatively, only when the calculated modulation rate exceeds one (modulation rate >100%), the modulation rate calculator **82** may output the calculated modulation rate or information that the calculated modulation rate exceeds one (modulation rate >100%) to the limiter adjuster **83**.

(128) The limiter adjuster **83** is provided as, for example, part of the modulation rate calculator **82** and is connected to the root mean square value calculator **51** through the modulation rate calculator **82** and also connected to the limiter **84** through the DC voltage operation range detector **85**. The limiter adjuster **83** is also connected to the storage **70** although wires are omitted in the drawing. The limiter adjuster **83** acquires information of the modulation rate from the modulation rate calculator **82** and acquires information of an operation range lower bound voltage set value v.sub.L3 and information of the rated value v.sub.R from the storage **70**.

(129) The limiter adjuster **83** calculates an operation range lower bound voltage operation value v.sub.L4 based on the operation range lower bound voltage set value v.sub.L3, the rated value v.sub.R, and the grid voltage root mean square value v.sub.RMS thus acquired. Note that the limiter adjuster **83** constantly performs the calculation and updates (calculates) the operation range lower bound voltage operation value v.sub.L4 whether the modulation rate exceeds one (modulation rate >100%) or does not exceed one (modulation rate ≤100%). Then, the limiter adjuster **83** outputs the calculated operation range lower bound voltage operation value v.sub.L4 to the limiter **84**. As a result, the modulation rate eventually becomes smaller than one.

(130) Note that the limiter adjuster **83** does not necessarily need to output the operation range lower bound voltage operation value v.sub.L4 to the limiter **84** when the operation range lower bound voltage operation value v.sub.L4 has no change as a result of the calculation. Alternatively, the limiter adjuster **83** may output the operation range lower bound voltage operation value v.sub.L4 to the limiter **84** constantly, at each predetermined interval, or at optional timings irrespective of the existence of variation in the modulation rate and the existence of variation in the operation range lower bound voltage operation value v.sub.L4.

(131) Note that the modulation rate calculator **82** and the limiter adjuster **83** are not limited to the configuration illustrated in FIG. 6, but the modulation rate calculator **82** and the limiter adjuster **83** may be an integrated component or completely separate components, or the limiter adjuster **83** may include the modulation rate calculator **82**.

(132) The limiter **84** is provided as, for example, part of the DC voltage operation range detector **85** and is connected to the limiter adjuster **83** through the modulation rate calculator **82**. The limiter **84** acquires information of the operation range lower bound voltage operation value v.sub.L4 from the limiter adjuster **83**. Note that the limiter **84** may acquire the information of the operation range lower bound voltage operation value v.sub.L4 from the limiter adjuster **83** only when the operation range lower bound voltage operation value v.sub.L4 has varied. Alternatively, the limiter **84** may acquire the information of the operation range lower bound voltage operation value v.sub.L4 from the limiter adjuster **83** constantly, at each predetermined interval, or at optional timings.

(133) When having acquired the information of the operation range lower bound voltage operation value v.sub.L4 from the limiter adjuster **83**, the limiter **84** updates the value of an operation range lower bound voltage v.sub.L4 set to the limiter **84**. Accordingly, the value of the operation range lower bound voltage v.sub.L4 is updated in accordance with the modulation rate. Note that the limiter **84** does not necessarily need to update the value of the operation range lower bound voltage v.sub.L4 when the operation range lower bound voltage operation value v.sub.L4 has no variation.

The limiter **84** sets the updated operation range lower bound voltage operation value $v_{\text{sub.L4}}$ as the present value of the operation range lower bound voltage $v_{\text{sub.L4}}$.

(134) The DC voltage operation range detector **85** is connected to the first voltage sensor **22** and the operation controller **86**. The DC voltage operation range detector **85** acquires the information of the DC voltage $v_{\text{sub.DC}}$ that is a value detected by the first voltage sensor **22**. The DC voltage operation range detector **85** compares the acquired information of the DC voltage $v_{\text{sub.DC}}$ and the present operation range lower bound voltage $v_{\text{sub.L4}}$ set to the limiter **84**. The DC voltage operation range detector **85** determines whether the acquired DC voltage $v_{\text{sub.DC}}$ is smaller than the operation range lower bound voltage $v_{\text{sub.L4}}$ set to the limiter **84** as a result of the comparison, and outputs information of a result of the determination to the operation controller **86**. When the acquired DC voltage $v_{\text{sub.DC}}$ is smaller than the operation range lower bound voltage $v_{\text{sub.L4}}$ set to the limiter **84** as a result of the comparison, the DC voltage operation range detector **85** outputs information that the DC voltage $v_{\text{sub.DC}}$ is smaller than the operation range lower bound voltage $v_{\text{sub.L4}}$ to the operation controller **86**. When the acquired DC voltage $v_{\text{sub.DC}}$ is a value equal to or larger than the operation range lower bound voltage $v_{\text{sub.L4}}$ set to the limiter **84** as a result of the comparison, the DC voltage operation range detector **85** outputs information that the DC voltage $v_{\text{sub.DC}}$ is equal to or larger than the operation range lower bound voltage $v_{\text{sub.L4}}$ to the operation controller **86**. Note that the DC voltage operation range detector **85** may output nothing to the operation controller **86** when the acquired DC voltage $v_{\text{sub.DC}}$ is equal to or larger than the operation range lower bound voltage $v_{\text{sub.L4}}$ set to the limiter **84** as a result of the comparison.

(135) The operation controller **86** is connected to the DC voltage operation range detector **85**, the failure detector **87**, and each component of the power conversion system **1A** although wires are omitted in the drawing. The operation controller **86** acquires information of whether the DC voltage $v_{\text{sub.DC}}$ is smaller than the operation range lower bound voltage $v_{\text{sub.L4}}$ from the DC voltage operation range detector **85**. The operation controller **86** also acquires, from the failure detector **87**, for example, information of whether failure or anomaly of the power conversion system **1A** is detected.

(136) When having acquired the information that the DC voltage $v_{\text{sub.DC}}$ is smaller than the operation range lower bound voltage $v_{\text{sub.L4}}$ or when having acquired the information that failure or anomaly of the power conversion system **1A** is detected, the operation controller **86** outputs, to each connected component, an operation instruction to stop the power conversion system **1A** for protection. For example, when having acquired these pieces of information, the operation controller **86** outputs an operation instruction to perform gate block (GB) of the inverter **13** and cuts (or stops) the power conversion system **1A** off the entire grid. Alternatively, for example, when having acquired these pieces of information, the operation controller **86** may output an operation instruction to display an alert (alarm) on, for example, a non-illustrated display unit or may output an operation instruction to open (cutoff) the DC switch **11** and the AC switch **16**.

(137) Note that when not having acquired the information that the DC voltage $v_{\text{sub.DC}}$ is smaller than the operation range lower bound voltage $v_{\text{sub.L4}}$ nor the information that failure or anomaly of the power conversion system **1A** is detected or when having acquired no information, the operation controller **86** outputs no operation instruction to each connected component. In this case, the inverter **13** is not subjected to gate block (GB) nor the like, but operation of the power conversion system **1A** continues.

(138) The failure detector **87** is connected to the first current sensor **21**, the first voltage sensor **22**, and the operation controller **86**. The failure detector **87** acquires information of the DC current $i_{\text{sub.DC}}$ that is a value detected by the first current sensor **21** and the information of the DC voltage $v_{\text{sub.DC}}$ that is a value detected by the first voltage sensor **22**. The failure detector **87** is utilized for a failure detection function and a display function. Note that, although not illustrated, the failure detector **87** is also included in the PV-PCS **1** illustrated in FIG. **1**.

(139) The failure detector **87** determines whether, for example, the acquired DC current $i_{\text{sub.DC}}$ or

DC voltage $v_{sub.DC}$ exceeds a predetermined threshold. When having determined that, for example, the acquired DC current $i_{sub.DC}$ or DC voltage $v_{sub.DC}$ exceeds the predetermined threshold, the failure detector **87**, determines that failure or anomaly has occurred to the power conversion system **1A**, and outputs information that failure or anomaly of the power conversion system **1A** is detected to the operation controller **86**. When having determined that, for example, the acquired DC current $i_{sub.DC}$ or DC voltage $v_{sub.DC}$ is equal to or smaller than the predetermined threshold, the failure detector **87** determines that no failure nor anomaly has occurred to the power conversion system **1A**, and may output information of the determination to the operation controller **86** or may output nothing.

(140) Note that although the operation controller **86** and the failure detector **87** are separated from each other in FIG. **6**, the operation controller **86** and the failure detector **87** may be integrated. In this case, when having detected failure or anomaly of the power conversion system **1A**, the failure detector **87** may output, to the failure detector **87** itself, an operation instruction to perform gate block (GB) of the inverter **13** or the like. Moreover, although the DC voltage operation range detector **85** and the operation controller **86** are separated from each other in FIG. **6**, the DC voltage operation range detector **85** and the operation controller **86** may be integrated. In this case, when having determined that the DC voltage $v_{sub.DC}$ is smaller than the operation range lower bound voltage $v_{sub.L4}$, the DC voltage operation range detector **85** may output, to the DC voltage operation range detector **85** itself, an operation instruction to perform gate block (GB) of the inverter **13** or the like.

(141) FIG. **7** is a diagram illustrating an example of the configuration and processing of the modulation rate calculator **82** and the limiter adjuster **83** illustrated in FIG. **6**. Note that, in FIG. **7**, any component identical or similar to a component illustrated in FIG. **2** is denoted by the same reference sign, and detailed description thereof is omitted or simplified.

(142) Although wires are omitted in the drawing, the modulation rate calculator **82** is connected to the first voltage sensor **22** and the root mean square value calculator **51** as described above with reference to FIG. **6**, acquires the information of the DC voltage $v_{sub.DC}$ from the first voltage sensor **22**, and acquires the information of the grid voltage root mean square value $v_{sub.RMS}$ from the root mean square value calculator **51**. The modulation rate calculator **82** calculates the modulation rate based on the information of the DC voltage $v_{sub.DC}$ and the information of the grid voltage root mean square value $v_{sub.RMS}$ thus acquired. The method of calculating the modulation rate is the same as the method described above with reference to FIG. **2**, and thus detailed description thereof will be omitted.

(143) Although wires are omitted in the drawing, the limiter adjuster **83** is connected to the root mean square value calculator **51**, the limiter **84**, and the storage **70** as described above with reference to FIG. **6**. The limiter adjuster **83** includes, for example, an operation range lower bound voltage set value acquirer **83a**, the grid voltage root mean square value acquirer **53b**, the rated value acquirer **53c**, the divider **53d**, the upper-lower bound limiter **53e**, the multiplier **53f**, and an operation range lower bound voltage operation value outputter **83g**.

(144) The operation range lower bound voltage set value acquirer **83a** is connected to the storage **70** and the multiplier **53f**. The operation range lower bound voltage set value acquirer **83a** acquires the operation range lower bound voltage set value $v_{sub.L3}$ from the storage **70** and outputs the acquired operation range lower bound voltage set value $v_{sub.L3}$ to the multiplier **53f**. Note that the operation range lower bound voltage set value $v_{sub.L3}$ is determined to be a predetermined value for each device as a specification, for example. In the example illustrated in FIG. **7**, the operation range lower bound voltage set value acquirer **83a** acquires 900 V as the operation range lower bound voltage set value $v_{sub.L3}$ from the storage **70** and outputs 900 V thus acquired to the multiplier **53f**.

(145) The operation range lower bound voltage operation value outputter **83g** is connected to the multiplier **53f** and the limiter **84**. The operation range lower bound voltage operation value

outputter **83g** acquires, from the multiplier **53f**, a value obtained by multiplying the operation range lower bound voltage set value $v_{\text{sub.L3}}$ by the value limited by the upper-lower bound limiter **53e** and outputs the acquired value as the operation range lower bound voltage operation value $v_{\text{sub.L4}}$ to the limiter **84**. In the example illustrated in FIG. 7, the operation range lower bound voltage operation value outputter **83g** acquires 945 V from the multiplier **53f** and outputs 945 V thus acquired as the operation range lower bound voltage operation value $v_{\text{sub.L4}}$ to the limiter **84**. (146) Through the above-described processing, when having acquired the information of the operation range lower bound voltage operation value $v_{\text{sub.L4}}$ from the limiter adjuster **83** (operation range lower bound voltage operation value outputter **83g**), the limiter **84** (refer to FIG. 6) updates the value of the operation range lower bound voltage $v_{\text{sub.L4}}$ set to the limiter **84**. Accordingly, it is possible to update the value of the operation range lower bound voltage $v_{\text{sub.L4}}$ in accordance with the modulation rate, thereby achieving a wider operation range on the DC side than in conventional cases and stably operating the power conversion system **1A** without overmodulation.

(147) Specifically, for example, the operation range lower bound voltage $v_{\text{sub.L4}}$ needs to be set in accordance with specifications of the energy storage system **4A** and the power conversion system **1A**. However, for example, when the operation range lower bound voltage $v_{\text{sub.L4}}$ is set to be higher than the specifications of the energy storage system **4A** and the power conversion system **1A** in advance to avoid overmodulation, the operation range on the DC side becomes narrow. In the example illustrated in FIG. 7, the operation range of the ESS-PCS **1A** is 900 V to 1300 V as illustrated at a lower part of FIG. 7, but when the lower bound voltage is increased with consideration on increase of the grid voltage, the operation range becomes, for example, 990 V to 1300 V and accordingly, the operation range on the DC side becomes narrow.

(148) However, according to the present embodiment, for example, it is possible to, in accordance with the modulation rate, increase the operation range lower bound voltage $v_{\text{sub.L4}}$ when the grid voltage root mean square value $v_{\text{sub.RMS}}$ has increased, and decrease the operation range lower bound voltage $v_{\text{sub.L4}}$ when the grid voltage root mean square value $v_{\text{sub.RMS}}$ has decreased. Accordingly, it is possible to update the value of the operation range lower bound voltage $v_{\text{sub.L4}}$ in accordance with the modulation rate, thereby achieving a wider operation range on the DC side than in conventional cases (without narrowing the operation range on the DC side) and stably operating the power conversion system **1A** without overmodulation.

Operation of Second Embodiment

(149) FIG. 8 is a flowchart illustrating an example of operation of the DC voltage operation range detector **85** and the operation controller **86** illustrated in FIG. 6. The flowchart illustrated in FIG. 8 is started when operation of the power conversion system **1A** (DC power supply from the energy storage system **4A**) is started.

(150) At step S21, the DC voltage operation range detector **85** acquires the information of the DC voltage $v_{\text{sub.DC}}$ that is a value detected by the first voltage sensor **22** from the first voltage sensor **22**.

(151) At step S22, when having acquired the information of the operation range lower bound voltage operation value $v_{\text{sub.L4}}$ from the limiter adjuster **83**, the limiter **84** of the DC voltage operation range detector **85** updates the value of the operation range lower bound voltage $v_{\text{sub.L4}}$ set to the limiter **84**. Note that the limiter **84** does not necessarily need to update the value of the operation range lower bound voltage $v_{\text{sub.L4}}$ when the operation range lower bound voltage operation value $v_{\text{sub.L4}}$ has no variation. The limiter **84** sets the present value of the operation range lower bound voltage $v_{\text{sub.L4}}$ to the updated operation range lower bound voltage operation value $v_{\text{sub.L4}}$. Note that the operation at step S22 is performed out of synchronization with the operation performed by the DC voltage operation range detector **85**. Thus, the operation at step S22 does not necessarily need to be performed at the timing indicated at step S22 but is performed at optional timings.

(152) At step S23, the DC voltage operation range detector **85** determines whether the DC voltage $v_{sub.DC}$ acquired at step S21 is smaller than the present operation range lower bound voltage $v_{sub.L4}$ set to the limiter **84**. When the DC voltage $v_{sub.DC}$ is smaller than the operation range lower bound voltage $v_{sub.L4}$ set to the limiter **84** (Yes), the DC voltage operation range detector **85** advances the processing to step S24. When the DC voltage $v_{sub.DC}$ is equal to or larger than the value of the operation range lower bound voltage $v_{sub.L4}$ set to the limiter **84** (No), the DC voltage operation range detector **85** advances the processing to step S25.

(153) At step S24, the operation controller **86** acquires the information that the DC voltage $v_{sub.DC}$ is smaller than the operation range lower bound voltage $v_{sub.L4}$ from the DC voltage operation range detector **85**. In this case, for example, the operation controller **86** outputs an operation instruction to perform gate block (GB) of the inverter **13** and cuts (or stops) the power conversion system **1A** off the entire grid. Then, the DC voltage operation range detector **85** and the operation controller **86** end the processing of the flowchart of FIG. **8**.

(154) At step S25, the operation controller **86** acquires, from the DC voltage operation range detector **85**, information that the DC voltage $v_{sub.DC}$ is equal to or larger than the operation range lower bound voltage $v_{sub.L4}$. Alternatively, the operation controller **86** acquires no information from the DC voltage operation range detector **85**. In this case, the operation controller **86** outputs no operation instruction to each connected component. Accordingly, the inverter **13** is not subjected to gate block (GB) nor the like, but operation of the inverter **13** (power conversion system **1A**) continues.

(155) At step S26, the DC voltage operation range detector **85** determines whether the operation of the ESS-PCS **1A** (DC power supply from the energy storage system **4A**) has ended. When having determined that the operation of the ESS-PCS **1A** (DC power supply from the energy storage system **4A**) has ended (Yes), the DC voltage operation range detector **85** ends the processing of the flowchart of FIG. **8**. When having determined that the operation of the ESS-PCS **1A** (DC power supply from the energy storage system **4A**) has not ended (No), the DC voltage operation range detector **85** returns the processing to step S21 and repeats the processing at steps S21 to S26.

Effects of Second Embodiment

(156) As described above, according to the second embodiment illustrated in FIGS. **6** to **8**, it is possible to update the operation range lower bound voltage $v_{sub.L4}$ in accordance with the modulation rate, thereby achieving a wider operation range on the DC side than in conventional cases and stably operating the power conversion system **1A** without overmodulation.

(157) In a case of the ESS-PCS **1A**, the DC voltage $v_{sub.DC}$ is used to monitor the device operation range, and according to the second embodiment illustrated in FIGS. **6** to **8**, the operation range lower bound voltage $v_{sub.L4}$ is variable in accordance with the modulation rate. Thus, it is possible to prevent the operation range on the DC side from becoming narrower than the specifications of the energy storage system **4A** and the power conversion system **1A** and stably operate the power conversion system **1A** without overmodulation.

(158) In addition, according to the second embodiment illustrated in FIGS. **6** to **8**, the update of the operation range lower bound voltage $v_{sub.L4}$ is performed by the modulation rate calculator **82** and the limiter adjuster **83**, separately from the DC voltage operation range detector **85**. In other words, the control to update the operation range lower bound voltage $v_{sub.L4}$ and the control by the DC voltage operation range detector **85** are independently performed in parallel by separate components. Thus, the DC voltage operation range detector **85** only needs to determine whether the DC voltage $v_{sub.DC}$ is smaller than the operation range lower bound voltage $v_{sub.L4}$ set to the limiter **84**, which simplifies the control by the DC voltage operation range detector **85**.

Furthermore, the DC voltage $v_{sub.DC}$ varies with the remaining charge amount of the energy storage system **4A** and the like, and accordingly, the control by the DC voltage operation range detector **85** is not affected by the control to update the operation range lower bound voltage $v_{sub.L4}$. Moreover, since the control to update the operation range lower bound voltage $v_{sub.L4}$

and the control by the DC voltage operation range detector **85** are performed out of synchronization, the control to update the operation range lower bound voltage $v_{sub.L4}$ can be swiftly performed at a correct timing in accordance with the modulation rate.

(159) <Example of Hardware Configuration>

(160) FIG. **9** is a conceptual diagram illustrating an example of a hardware configuration of the processing circuit **90** included in the control devices **30** and **30A** in the embodiments illustrated in FIGS. **1** to **8**. Each above-described function is implemented by the processing circuit **90**. In an aspect, the processing circuit **90** includes at least one processor **91** and at least one memory **92**. In another aspect, the processing circuit **90** includes at least one dedicated hardware component **93**.

(161) When the processing circuit **90** includes the processor **91** and the memory **92**, each function is implemented by software, firmware, or combination of software and firmware. At least one of the software and the firmware is written as a computer program. The at least one of the software and the firmware is stored in the memory **92**. The processor **91** implements each function by reading and executing the computer program stored in the memory **92**.

(162) When the processing circuit **90** includes the dedicated hardware component **93**, the processing circuit **90** is, for example, a single circuit, a composite circuit, a programmed processor, or combination thereof. Each function is implemented by the processing circuit **90**.

(163) Some or all functions of the control devices **30** and **30A** may be implemented by hardware or may be implemented by a computer program executed by a processor. In other words, the control devices **30** and **30A** may be implemented by a computer and a computer program, and the computer program may be stored in a storage medium or provided through a network.

Supplementary Notes of Embodiments

(164) As described above, the embodiments illustrated in FIGS. **1** to **9** are the first embodiment illustrated in FIGS. **1** to **5** and the second embodiment illustrated in FIGS. **6** to **8**, but these embodiments may be combined in series or in parallel. Specifically, the power conversion system **1** or **1A** may include both the solar panel **4** and the energy storage system **4A** or may be applied to both. The combined embodiments can achieve the same effects as those of the embodiments not combined.

(165) In the embodiments illustrated in FIGS. **1** to **9**, the power conversion systems **1** and **1A** and the control devices **30** and **30A** included therein are exemplarily described as aspects of the present disclosure, but the present disclosure is not limited thereto. The present disclosure may be implemented as a control method of performing the processing steps at components of the control devices **30** and **30A**.

(166) Moreover, the present disclosure may be implemented as a control program configured to cause a computer to execute the processing steps at components of the control devices **30** and **30A**.

(167) Furthermore, the present disclosure may be implemented as a storage medium (non-transitory computer-readable medium) in which the control program is stored. The control program may be stored and distributed in, for example, a removable media such as a compact disc (CD), a digital versatile disc (DVD), or a universal serial bus (USB) memory. Note that the control program may be uploaded onto a network through a non-illustrated network interface or the like included in the control devices **30** and **30A** and may be downloaded from the network and stored in the storage **70** or the like.

(168) The features and advantages of the embodiments should be clarified through the above detailed description. The claims include the features and advantages of embodiments as described above without departing from the spirit and scope thereof. Various modifications and changes of the present invention should be easily thought of by a person with ordinary skill in the art. Thus, the range of inventive embodiments is not intended to be limited to those described above but may include appropriate modifications and equivalents in a range disclosed in the above-described embodiments.

REFERENCE SIGNS LIST

(169) **1** power conversion system (PV-PCS, PCS) **1** DC voltage command value **1A** power conversion system (ESS-PCS, PCS) **2** DC bus **2** DC voltage command value **3** AC circuit **4** solar panel (solar battery panel, PV panel) **4A** energy storage system (ESS) **5** AC power grid (power grid, grid) **11** DC switch (DC breaker, DC switch) **12** DC capacitor (DC capacitor) **13** inverter (inverter circuit) **14** AC reactor (AC reactor) **15** AC capacitor (AC capacitor) **15a** bifurcation point **16** AC switch (AC breaker, AC switch) **21** first current sensor **22** first voltage sensor **23** second current sensor **24** second voltage sensor **25** third current sensor **30**, **30A** control device **41** PLL controller **42** first converter **43** second converter **44** power controller **51** root mean square value calculator (RMS calculator) **52** modulation rate calculator **53** limiter adjuster **53a** lower bound limiter set value acquirer **53b** grid voltage root mean square value acquirer **53c** rated value acquirer **53d** divider **53e** upper-lower bound limiter **53f** multiplier **53g** lower bound limiter operation value outputter **54** limiter **55** maximum power point tracking controller (MPPT controller) **56** first subtractor **57** DC voltage controller **61** first adder **62** third converter **63** second subtractor **64** current controller **65** pulse width modulation controller (PWM controller, inverter controller) **70** storage **82** modulation rate calculator **83** limiter adjuster **83a** operation range lower bound voltage set value acquirer **83g** operation range lower bound voltage operation value outputter **84** limiter **85** DC voltage operation range detector **86** operation controller **87** failure detector **90** processing circuit **91** processor **92** memory **93** hardware component a straight line a' dashed line b thin straight line b' thin dashed line c thick dashed line c' dashed line i*.sub.d d-axial current command value i*.sub.q q-axial current command value i*.sub.AC AC current command value i.sub.AC inverter output current (current) i.sub.DC DC current (current) i.sub.Grid AC current (grid current, current) P, P.sub.1, P.sub.1A, P.sub.1B, P.sub.2, P.sub.2A, P.sub.2B power P.sub.mpp, P.sub.mppA, P.sub.mppB optimum operation point P.sub.VA, P.sub.VB PV curve t, t.sub.1 to t.sub.14 time point v'*.sub.DC DC voltage command value (first DC voltage command value) v*.sub.DC DC voltage command value (second DC voltage command value) V voltage v.sub.DC DC voltage (voltage) v.sub.Grid grid voltage (voltage) v.sub.L DC voltage command lower bound limiter v.sub.L1 lower bound limiter set value v.sub.L2 lower bound limiter operation value v.sub.L3 operation range lower bound voltage set value v.sub.L4 operation range lower bound voltage operation value v.sub.L4 operation range lower bound voltage V.sub.OCA, V.sub.OCB open voltage v.sub.R rated value v.sub.RMS grid voltage root mean square value θ reference phase

Claims

1. A control device in a photovoltaics power conversion system connected to a solar panel on a DC side, the control device comprising: a modulation rate calculator configured to calculate a modulation rate based on a DC voltage on a DC side of the power conversion system and a grid voltage root mean square value on an AC side of the power conversion system; a limiter adjuster configured to calculate a lower bound limiter operation value based on a predetermined lower bound limiter set value, the grid voltage root mean square value, and a predetermined rated value in accordance with the modulation rate calculated by the modulation rate calculator; a limiter configured to update a DC voltage command lower bound limiter based on the lower bound limiter operation value calculated by the limiter adjuster; a maximum power point tracking controller configured to calculate a first DC voltage command value by performing maximum power point tracking control based on DC power generated by the solar panel and determine a second DC voltage command value based on the calculated first DC voltage command value and the DC voltage command lower bound limiter set to the limiter; and an inverter controller configured to control an inverter of the power conversion system based on the second DC voltage command value determined by the maximum power point tracking controller.
2. The control device according to claim 1, wherein when the lower bound limiter operation value is calculated by the limiter adjuster, the limiter updates the DC voltage command lower bound

limiter based on the lower bound limiter operation value calculated by the limiter adjuster while being out of synchronization with the maximum power point tracking control by the maximum power point tracking controller.

3. The control device according to claim 1, wherein the limiter adjuster includes an upper-lower bound limiter configured to limit upper and lower bound values of the lower bound limiter operation value so that the lower bound limiter operation value does not become smaller than the predetermined lower bound limiter set value at least.

4. The control device according to claim 1, wherein the maximum power point tracking controller determines the value of the DC voltage command lower bound limiter to be the second DC voltage command value when the calculated first DC voltage command value is smaller than the value of the DC voltage command lower bound limiter set to the limiter, and the maximum power point tracking controller determines the first DC voltage command value to be the second DC voltage command value when the calculated first DC voltage command value is equal to or larger than the value of the DC voltage command lower bound limiter set to the limiter.

5. A photovoltaics power conversion system connected to a solar panel at which power generation is performed by sunlight, the photovoltaics power conversion system comprising: an inverter configured to convert DC power generated by the solar panel into AC power; and the control device according to claim 1.

6. A control device in an energy-storage-system power conversion system connected to an energy storage system on a DC side, the control device comprising: a modulation rate calculator configured to calculate a modulation rate based on a DC voltage on a DC side of the power conversion system and a grid voltage root mean square value on an AC side of the power conversion system; a limiter adjuster configured to calculate an operation range lower bound voltage operation value based on a predetermined operation range lower bound voltage set value, the grid voltage root mean square value, and a predetermined rated value in accordance with the modulation rate calculated by the modulation rate calculator; a limiter configured to update the operation range lower bound voltage based on the operation range lower bound voltage operation value calculated by the limiter adjuster; a DC voltage operation range detector configured to compare the DC voltage on the DC side of the power conversion system and the operation range lower bound voltage set to the limiter and determine whether the DC voltage is smaller than the operation range lower bound voltage; and an operation controller configured to stop an inverter of the power conversion system when it is determined by the DC voltage operation range detector that the DC voltage is smaller than the operation range lower bound voltage.

7. The control device according to claim 6, wherein when the operation range lower bound voltage operation value is calculated by the limiter adjuster, the limiter updates the operation range lower bound voltage based on the operation range lower bound voltage operation value calculated by the limiter adjuster while being out of synchronization with control by the DC voltage operation range detector.

8. The control device according to claim 6, wherein the limiter adjuster includes an upper-lower bound limiter configured to limit upper and lower bound values of the operation range lower bound voltage operation value so that the operation range lower bound voltage operation value does not become smaller than the predetermined operation range lower bound voltage set value at least.

9. An energy-storage-system power conversion system connected to an energy storage system capable of charging and discharging electric power, the energy-storage-system power conversion system comprising: an inverter configured to convert DC power supplied from the energy storage system into AC power; and the control device according to claim 6.
